

AYUSH

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OBJECTIVE

Early-career engineer with hands-on experience in Python-based data pipelines, ML model training, and feature engineering. Seeking an AI/ML Engineering role where I can contribute to production-grade data and ML systems, deepen my understanding of real-world model constraints, and grow into an AI/ML Platform Engineer.

EDUCATION

Thapar Institute of Engineering & Technology
B.E. in Computer Science & Engineering

– 2027
Patiala, India

EXPERIENCE

Research Intern — Experiential Learning Centre, Thapar University

Jun 2025 – Jul 2025

Built an end-to-end speech-emotion recognition pipeline using Python — from raw data collection to model deployment-ready output.

- Collected and preprocessed large-scale Bengali speech data from talk shows; applied noise reduction, audiosegmentation, and WAV-standard formatting to prepare a clean training dataset.
- Designed a feature engineering pipeline using MFCC-based extraction, transforming raw audio signals into structured numerical features suitable for ML model input.
- Trained and benchmarked multiple classification models (Random Forest, SVM, CatBoost); optimised CatBoost hyperparameters to achieve **82% accuracy** on the validation set.
- Developed modular, reproducible Python scripts for each pipeline stage (ingestion, preprocessing, feature extraction, training, evaluation) — structured for maintainability and future scaling.

PROJECTS

Solar Panel Efficiency Prediction

Python, XGBoost, LightGBM,
CatBoost, Optuna, SHAP Regression

pipeline for real-world solar energy data.

- Built a supervised regression pipeline using GPU-optimised gradient boosting models (XGBoost, LightGBM, CatBoost), achieving ~90% prediction accuracy on held-out test data.
- Engineered advanced features — polynomial transforms, weather-irradiance interaction terms, and sensor metadata — directly improving model stability across cross-validation folds.
- Used Optuna for automated hyperparameter tuning and SHAP to identify feature importance, reducing model variance and producing interpretable outputs for non-technical stakeholders.
- Structured the codebase as a modular ML workflow with versioned experiments, error handling, and scalable data-loading — designed with production readiness in mind.

AI-Driven Resume Ranking System

Python, NLP, spaCy, Transformers,
SQL, FastAPI

Automated screening engine using NLP + structured data pipelines.

- Built a document processing pipeline that extracts structured information (skills, experience, projects) from unstructured resume text using NLP preprocessing and transformer-based embeddings.
- Designed a rule-augmented scoring algorithm for candidate-role matching, reducing manual screening time by over 70% for bulk application batches.
- Implemented an SQL-backed data storage layer to persist parsed resume data, enabling fast retrieval and analytics — demonstrating practical database design for production workflows.

- Exposed pipeline functionality via RESTful APIs (FastAPI), enabling integration with hiring dashboards and ApplicantTracking Systems (ATS).

TECHNICAL SKILLS

Languages	Python, C/C++, SQL
ML Libraries	Scikit-learn, XGBoost, LightGBM, CatBoost, PyTorch, Transformers
Data & Pipelines	Pandas, MFCC/audio processing, feature engineering, model evaluation
Databases	MySQL, PostgreSQL, MongoDB
APIs & Tools	FastAPI, REST APIs, Git/GitHub, Linux, VS Code

ACHIEVEMENTS

Zelestra × AWS ML Ascend Challenge (2025)

Ranked **Top 6% globally** — 63rd out of 1,000+ teams — for an end-to-end ML solution built and evaluated under competition constraints.

Shell.ai Hackathon 2025 (20-day)

Secured **21st Global Rank** by building a high-performance hybrid ML pipeline to predict sustainable fuel blend properties under real-world data constraints.

Smart India Hackathon 2025 — Internal Round

Cleared SIH 2025 internal hackathon (Problem SIH25108), selected from 256 participating teams.